

Just as the components of a vector change when we change the [basis](#) of the vector space, the components of a tensor also change under such a transformation. Each type of tensor comes equipped with a *transformation law* that details how the components of the tensor respond to a [change of basis](#). The components of a vector can respond in two distinct ways to a [change of basis](#) (see [covariance and contravariance of vectors](#)), where the new [basis vectors](#) $\hat{\mathbf{e}}_i$ are expressed in terms of the old basis vectors $\mathbf{e}_j$ as,

$$\hat{\mathbf{e}}_i = \sum_{j=1}^n \mathbf{e}_j R_i^j = \mathbf{e}_j R_i^j.$$

Here R_i^j are the entries of the change of basis matrix, and in the rightmost expression the [summation](#) sign was suppressed: this is the [Einstein summation convention](#), which will be used throughout this article.¹ The components v^i of a column vector $\mathbf{v}$ transform with the [inverse](#) of the matrix R ,

$$\hat{v}^i = (R^{-1})^i_j v^j,$$

where the hat denotes the components in the new basis. This is called a *contravariant* transformation law, because the vector components transform by the *inverse* of the change of basis. In contrast, the components, w_i , of a covector (or row vector), $\mathbf{w}$, transform with the matrix R itself,

$$\hat{w}_i = w_j R_i^j.$$

¹ The Einstein summation convention, in brief, requires the sum to be taken over all values of the index whenever the same symbol appears as a subscript and superscript in the same term. For example, under this convention $B_i C^i = B_1 C^1 + B_2 C^2 + \dots B_n C^n$